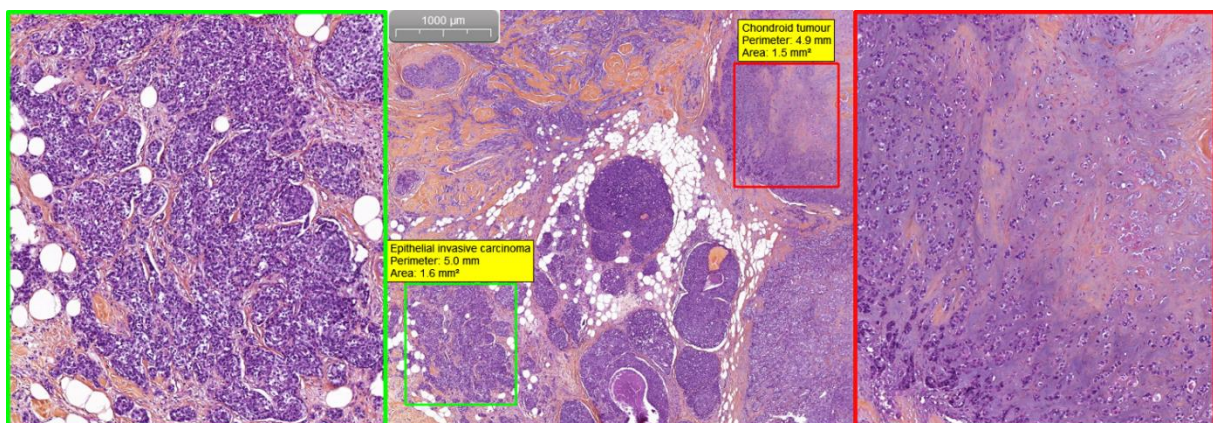


## 4) Project

### Evolutionary determinants and trajectories of metaplastic breast carcinomas

#### Introduction

Triple-negative breast cancers (TNBCs) face a lack of therapeutic options and high mortality rates<sup>1</sup>. **Metaplastic breast carcinomas (MpBCs) are a rare subtype of TNBCs, characterized by high cellular plasticity** and the presence of at least one transdifferentiated tumor compartment<sup>2,3</sup> (**Figure 1**). These compartments can be of various non-epithelial types, whose presence is determined exclusively through histological analysis. In addition, the mechanisms underlying their emergence and phenotypic switch remain unexplained. Cellular plasticity, which allows for these dynamic phenotypic changes, may also contribute to therapeutic resistance through non-genetic mechanisms<sup>4,5</sup>. Due to their rarity and complexity, the biology of MpBCs remains poorly understood, and patients are therefore still inadequately managed.



**Figure 1 – Example of a mixed MpBC sample.**

*Within the same tumor, an area of invasive epithelial breast carcinoma is visible on the left (magnified in the green box), and an area of chondroid transdifferentiation is visible on the right (magnified in the red box).*

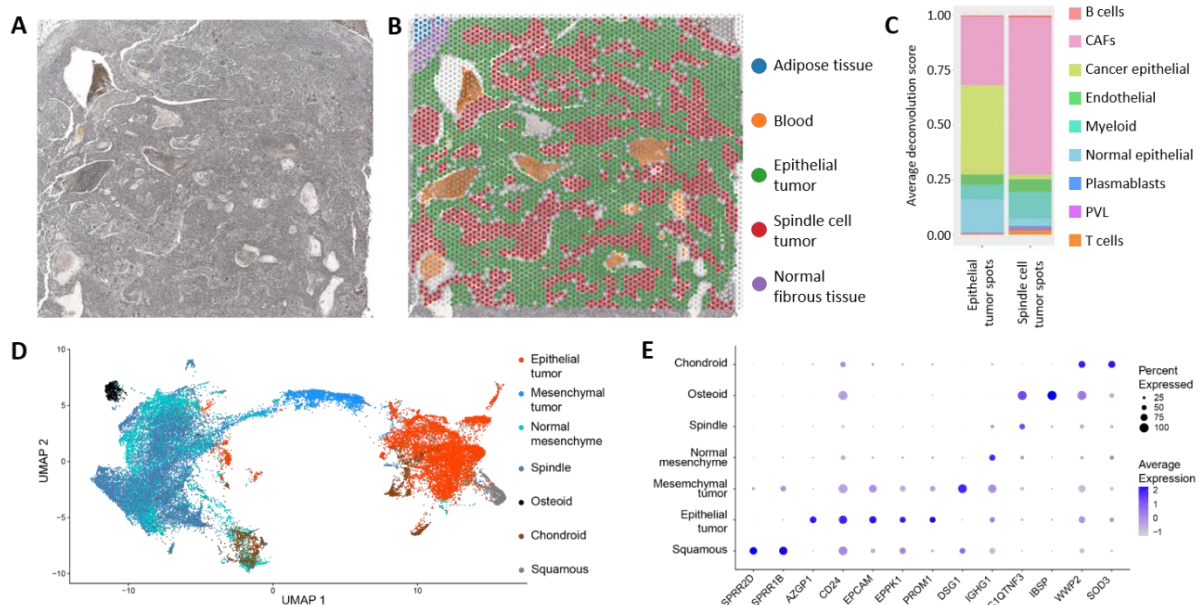
The objectives of this project are to **better understand the emergence of transdifferentiated compartments in MpBCs and to provide molecular diagnostic options for these tumors**. I will focus on mixed-histology MpBCs, in which phenotypically distinct compartment pairs can be studied, using high-precision multi-omics technologies: spatial transcriptomics and single-cell RNA sequencing (scRNA-seq); exome and methylome sequencing after microdissection. This approach will help uncover the origin and genetic/non-genetic mechanisms underlying **four different types of transdifferentiation: spindle cell, squamous, chondroid, and osteoid**. Fifteen mixed MpBC samples have already been analyzed using Visium spatial transcriptomics, and 15 others are currently being processed. The availability of orthogonal scRNA-seq data will enable more precise identification of deregulated genes and signaling pathways in transdifferentiated MpBC cells—both in comparison to classical epithelial cancer cells and to normal mesenchymal cells they resemble histologically. This will lay the groundwork for identifying new molecular diagnostic and classification solutions, as well as therapeutic opportunities that will be validated by the team's collaborators.

To achieve its goals, the project is structured around three specific aims:

- 1) **Creation** of a single-cell and spatial resolution transcriptomic atlas dedicated to MpBC.
- 2) **Characterization** of the genetic and non-genetic determinants of transdifferentiation in MpBC.
- 3) **Identification** of subtype-specific expression markers and signaling pathways.

## Creation of a Single-Cell and Spatial Transcriptomic Atlas for MpBC

My project will enable a comprehensive characterization of the cellular and molecular features underlying MpBC heterogeneity. Despite the existence of multiple transcriptomic datasets on inter- and intra-tumoral heterogeneity in breast cancers, none are specific to MpBC. As shown by our preliminary data, this limits the ability of existing datasets to accurately deconvolute spatial transcriptomics data. Due to a lack of appropriate reference data, algorithms incorrectly attribute RNA profiles from Visium “spots” annotated as spindle to fibroblasts (**Figure 2A-C**). Despite the existence of expression signatures to detect the epithelial-to-mesenchymal transition (EMT), we still lack dedicated data on transdifferentiated cells in these breast cancers. The goal is therefore to **create an atlas, i.e. a precise and appropriate reference dataset—for decoding tumor cell heterogeneity in MpBC**, using single-cell RNA sequencing (scRNA-seq) combined with spatial transcriptomics.



**Figure 2 – Preliminary spatial transcriptomic data.**

A) Pre-Visium microscopic image of a mixed MpBC sample with spindle cells.

B) Visium spots (~10 cells) annotated.

C) Deconvolution scores of the different cell types contributing to the global transcriptomic signal of epithelial and spindle tumor spots.

D) UMAP projection of annotated spots from 15 mixed MpBC samples. Although subtype-specific clusters can be observed, the separation is imperfect due to Visium spots potentially containing mixtures of cell types.

E) Putative subtype-specific expression markers identified from preliminary analyses of annotated Visium spots. Higher-resolution single-cell data may yield even more relevant biomarkers.

As in previous work by the team<sup>6</sup> and in my preliminary analyses, Visium slides will be annotated individually using a hybrid approach, combining K-means clustering with manual annotation by a trained pathologist. The **snPATHO-seq** protocol<sup>7</sup>, jointly developed by the single-cell platform at the Centre Léon Bérard (CLB), will be implemented to obtain single-nuclei RNA sequencing (snRNA-seq) data—a scRNA-seq variant better suited to FFPE samples used here. This method allows both **cell type annotation and copy number alteration (CNA) profiling**, with precision similar to cell-based methods. We will use it to create a single-cell expression dataset covering all rare MpBC cell types. For each of

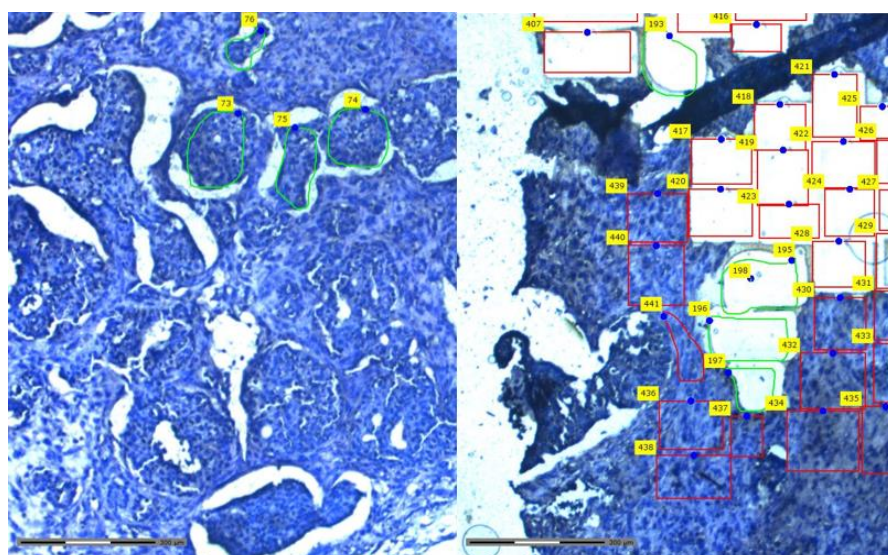


the 4 types of transdifferentiation studied, I will select the 5 best spatially profiled samples (n=20), based on Visium quality control metrics and the clarity of separation between compartments.

A limitation of snRNA-seq is the lack of cross-referenced datasets for accurate annotation, especially for the rare subtypes studied here. However, spatial transcriptomic data and pathologist annotations will help overcome this. Marker expression profiles from snRNA-seq clusters will be cross-examined against dimensionally-reduced projections of our spatial transcriptomics data (**Figure 2D-E**), to confirm their match with specific transdifferentiation states.

### Characterization of the genetic and non-genetic determinants of transdifferentiation in MpBC

To better understand the evolutionary trajectories of the 20 selected mixed MpBC samples, the team and its collaborators (Dr. M. Ouzounova, CRCL) are microdissecting the samples using a laser capture microscope (Zeiss PALM MicroBeam, CLB research pathology platform). This will allow **precise identification of genetic features in both compartments of each sample (Figure 3)**.



**Figure 3 – Laser capture microdissection.**

*Selection of epithelial (green) and mesenchymal (red) areas in a case of mixed MpBC. Cresyl violet staining. Microdissected fragments from the two compartments are collected into separate tubes, and 8–10 serial sections are microdissected for each sample. Replicates from the same compartment are then pooled per sample for compartment-specific DNA extraction.*

Somatic genetic alterations—point mutations, CNAs, and indels—will be identified through whole-exome sequencing (Twist Biosciences 2.0 kit) at 200x coverage. The Mutect2 software<sup>8</sup> will be used for point mutation and indel analyses. In the absence of a normal control (blood or healthy tissue), paired compartments will serve as each other's reference to identify compartment-specific mutations. A mutation will be considered specific if present (allele frequency >15%) in one compartment and entirely absent in the other. Mutations listed as “drivers” in the COSMIC database and with >15% allele frequency in both compartments will be considered clonal (present in the original cancer cell). PureCN<sup>9</sup> will be used to identify compartment-specific CNAs, using a pool of normal samples already sequenced by the team as reference.

To explore potential non-genetic determinants of transdifferentiation, deconvolution analyses will be performed using RCTD<sup>10</sup>, combining the existing breast cancer atlas data<sup>11</sup> and our dedicated MpBC atlas. **Differences in tumor microenvironment (TME) composition will be evaluated** by comparing spot compositions for each (trans)differentiation type, using ANOVA. To investigate whether external factors drive transdifferentiation, we will specifically analyze the composition of inner and outer borders of spatially segregated clusters (>25 spots) associated with transdifferentiated cells. Finally, using ligand-receptor databases<sup>12–14</sup> and the SpatialDM<sup>15</sup> tool, we will focus on **identifying ligand-receptor pairs likely to indicate interactions** between non-cancerous cells and transdifferentiated cancer cells.

## Identification of subtype-specific expression markers and signaling pathways

Despite the existence of transcriptomic signatures like the claudin-low subtype<sup>16,17</sup>, associated with MpBC<sup>18</sup>, the team's previous work has shown that **molecular approaches for MpBC identification remain inadequate**. This is especially due to similarities between mesenchymal stroma and transdifferentiated cells<sup>6</sup>. Differential expression analyses will therefore be conducted to identify specific biomarkers for each compartment type, based on snRNA-seq data (rather than the less accurate Visium data). Normalization and integration will be performed using Seurat. Two different methods will be implemented and compared to identify compartment-specific markers: 1) Seurat's FindAllMarkers function, and 2) MAST<sup>19</sup>, which uses a Hurdle model to correct for fixed and random effects, thereby efficiently addressing batch and sample biases. This approach ensures that pathological annotations from our Visium data provide reliable orthogonal information for single-cell classification, which in turn enables **more precise biomarker identification**.

We will also seek to **identify key regulators** potentially driving differential expression in each MpBC subtype. Using the SCENIC tool<sup>20</sup>, we will identify gene regulatory networks specific to transdifferentiation subtypes and the transcription factors responsible for the observed expression profiles. This should narrow down the list of candidate genes for in-depth study and overcome the challenge of identifying relevant but lowly expressed genes. Finally, we will combine differential expression analyses performed across similar subtypes and within individual mixed-histology tumors. This will ensure that the selected genes are representative of most or all samples within each subtype, rather than being statistically significant but only relevant to a subset of samples.

## Impact, feasibility, and perspectives

MpBC is a rare subtype of breast cancer that is still poorly managed. By addressing this issue, my project will benefit both researchers and healthcare professionals working on MpBC, as well as the affected patients. I will create and share a transdifferentiation atlas of MpBC at single-cell resolution with paired spatial resolution. This will **fill a significant knowledge gap** and serve as a valuable resource for future research.

My PhD project is part of a **national initiative** involving the Cancer Research Center of Lyon and the Centre Léon Bérard (Lyon), the Institut Curie (Paris), the Bergonié Institute (Bordeaux), and the Cancer Research and Immunology Center (Nantes). This initiative aims to improve the management of patients with metastatic breast cancer and to develop PDX and organoid models for studying transdifferentiation in aggressive breast cancer forms. This national effort is **supported by Ruban Rose, Ligue Régionale Contre le Cancer, and InterSIRIC funding**, which has enabled the administrative and logistical setup (legal agreements and material transfer between institutions), as well as the production of the multi-omics data I will analyze.

Finally, this national collaboration will enable the clinical and functional validation of the biomarkers I will identify. In collaboration with Institut Curie (Dr. V. Cockenpot), the specificity of biomarkers for each transdifferentiation type will be **validated by immunohistochemistry on an independent cohort**. This will assess their relevance for diagnostic and classification purposes in molecular analyses. Lastly, thanks to the InterSIRIC partnership, the functional impact of these markers will be studied through **genetic and pharmacological perturbations in two patient-derived MpBC models**: mouse xenografts (Dr. E. Marangoni, Institut Curie) and organoids (Dr. V. Guen, Cancer Research and Immunology Center of Nantes).

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